

Finger Training for Strength and Endurance

There's no such thing as too much strength.

—Wolfgang Güllich

This chapter on training grip strength and forearm muscle endurance will likely be the most referred to in the book and understandably so, because the fingers are usually the epicenter of fatigue and failure in rock climbing. Of course, the self-aware climber will always analyze whether failure on a given route was actually the result of an underlying problem in sequencing, technique and tactics, or mental control. This distinction aside, gaining more finger strength and endurance is obviously central to taking your climbing to the next level. Read and appropriately utilize the exercises in this chapter, and you will undoubtedly improve in this area—perhaps massively—and crush at a higher level in the months and years to come!

While beginning climbers will get stronger simply by climbing a few days per week, more experienced climbers need to engage in some targeted training of the finger-controlling muscles of the forearms. As detailed in chapter 5, effective training must zero in on a specific functional aspect and exercise the fingers flexors in a precise way to target the related fiber type and energy system. Accordingly, this chapter is broken into four sections that detail exercises for training maximum grip strength, contact strength (power), strength-endurance, and local aerobic endurance.

The exercises in each section are organized roughly by relative difficulty, beginning with

the easier exercises and ending with the more advanced. Exercise difficulty and the appropriateness of an exercise for a given ability level is a most vital distinction. A novice climber could be injured in attempting an advanced exercise such as feet-off campusing, whereas an advanced climber would gain little outside a warm-up by performing an entry-level exercise. Consult table 8.1 for a classification of the exercises that will help guide you in proper selection.

As you engage in finger training over the months and years to come, it's important to recognize that strength and endurance gains will not come in a steady, linear fashion. While initial gains will be realized quickly, long-term improvement will be harder to recognize as apparent performance plateaus develop. Breaking through these frustrating plateaus demands training both smarter and harder. You just can't do more of everything (a trap that many passionate climbers fall into) hoping to make a breakthrough; instead you must identify your limiting constraint and make it the focus of your training. For most climbers, long-term gains demand a more polarized approach that focuses equally on increasing absolute strength (anaerobic alactic) and developing greater aerobic power and endurance. Ironically, many climbers do just the opposite in worshipping the false idol of “the pump” (anaerobic lactic training) as the ultimate validation of an effective workout. Certainly you must do some training of anaerobic capacity, but as presented in chapter 5, this is the least trainable

Table 8.1 Classification of Finger/Forearm Training Exercises

	Exercise	Beginner	Intermediate	Advanced
Maximum Strength (anaerobic alactic)	Bouldering	X	X	X
	Hypergravity Bouldering		X	X
	Fingerboard "Minimum Edge" Hangs		X	X
	Fingerboard "Maximum Weight" Hangs			X
	HIT System with Max Weight			X
Contact Strength & Power (anaerobic alactic)	One-Arm Lunging (feet on)		X	X
	Campus Laddering (small holds)		X	X
	Campus Board "Bumps" (small holds)		X	X
	Campus Board "Switch Hands"			X
	Campus Board Double Dynos			X
Strength-Endurance (glycolytic/ anaerobic lactic)	Fingerboard Repeaters (short duration)	X	X	X
	Fingerboard Repeaters (long duration)		X	X
	Bouldering 4x4s	X	X	X
	HIT System—Strength-Endurance Protocol		X	X
	System Wall "Repeaters"		X	X
Local Aerobic Endurance (aerobic/oxidative)	ARC Traverses ("Recovery Climbing")	X	X	X
	Fingerboard Moving Hangs		X	X
	Campus Board "Hand Play" (feet-on)		X	X
	Threshold Treadwall Intervals		X	X
	Route Climbing Intervals	X	X	X

energy system and the most likely to lead to over-training (and negative results).

A final comment from Coach Hörst before we dig into the exciting exercise details: The number one rule of finger training is *Don't get injured!* While this rule is presumably self-evident, a surprising number of enthusiastic climbers get injured (fingers, elbows, shoulders) by training too much or in attempting advanced exercises for which they are not ready. Here are some important finger-training guidelines that will reduce your risk of injury:

- Train wrist and shoulder stability (see chapter 6) before and concurrent with targeted finger exercise training.
- Always perform a progressive warm-up that gradually builds from easy full-body activity to difficult, climbing-specific exercises.
- Make a conscious effort to avoid the most painful and stressful holds (such as a sharp, small-radius edge or a pocket that feels tweaky). No move or boulder problem is worth getting injured over!
- Limit training redundancy by using a few different exercises each session. Don't just train for an hour on a single apparatus (such as the fingerboard or campus board).
- Don't try to train everything—maximum strength, contact strength, strength-endurance, and local aerobic endurance—in a single session. Always train with a specific focus.
- Immediately end an exercise at the first sign of any tendon or joint pain. Stop training for a few minutes (or days) to assess the cause of the pang and severity of the potential injury.
- Rest more than you think you need to—your muscles, tendons, and precious digits will thank you! Three to five days of aggregate training and climbing per week is enough, and any more tempts injury.

Training Maximum Strength

“There's no such thing as too much strength” is a now iconic quote of the late, great Wolfgang Güllich, and it's a good prime directive for your training. But what's the best way to train for greater finger strength?

According to the principle of specificity, effective finger training must necessarily target the neuromuscular system in ways that are specific to climbing. Therefore, fingerboard hangs, cam-pusing, and bouldering are all valid ways to train finger strength for climbing, whereas a hand-held squeeze device is a waste of time as anything other than a warm-up exercise. The degree to which a given exercise will produce gains in functional grip strength for climbing can be estimated by considering the following requirements—the more of these requirements that are met, the more effective the exercise will be at producing usable gains in climbing grip strength.

1. The exercise must be near maximum intensity throughout the entire set. In climbing-specific finger training you can increase intensity by decreasing hold size or adding weight. Of course, there's a definite limit to how small a hold you can train on comfortably—holds smaller than about 10mm ($\frac{3}{8}$ inch) deep are often painful to train on. A popular alternative, then, is to use a somewhat larger hold (14 to 20mm or $\frac{5}{8}$ to $\frac{3}{4}$ inch is ideal) and add weight to your body to increase intensity.

2. The exercise must produce muscular failure in less than twelve seconds. As detailed in chapter 5, maximal exercise is powered by the anaerobic alactic energy system, so this energy system must be the training focus. Exercising any longer than about twelve seconds per set trains strength-endurance (the anaerobic lactic energy system), not maximum strength—this is a critical distinction!

3. The exercise must be specific to climbing grip positions and movements. Obviously

Self-Testing of Finger Strength and Endurance

As a climbing coach with deep background in science, I'm enamored with fitness testing—that is, using a specific exercise with a fixed protocol to measure a physical capability and track changes over the long term. Following are three simple, yet effective methods of testing limit strength and strength-endurance of your forearm flexors (primary limiting constraints for many climbers). Perform these tests every few months (or training cycle) to track your gains and modify your program appropriately, year over year. See chapter 9 for six tests to track your pull-muscle limit strength, power, and endurance.

Finger strength test: Determine the maximum amount of weight you can add to your body (via weights hanging from the belay loop of a harness) and hang for five seconds from a fingerboard 10mm (one-half finger pad size) edge using a half-crimp grip (no thumb lock). Finding the maximum weight will take a few attempts. Be sure to perform an extensive warm up beforehand, and rest a minimum of three minutes between hangs. In case you're wondering, elite climbers can typically hang five seconds with an added weight of at least one third of their body weight—this is a good long-term (multi-year) goal to train toward.

(**Note:** Small differences in hold size will lead to large differences in weight you can hold, so it's important to test on the same 10mm hold each time.)

Finger endurance test #1: Determine how long you can endure through series of repeated five-second hangs, with exactly five seconds of rest between each hang, on a 10mm fingerboard edge. After a thorough warm up, have a training partner direct you through a series of precisely timed five-second intervals—rests between hangs must never exceed five seconds. The test ends when you can no longer hang for five seconds. Use only half-crimp, open-crimp and/or open-hand grips, and strive to maintain good fingerboard technique throughout. This test closely simulates how the forearm flexors work (intermittent blood-flow occlusion and recovery) when climbing a difficult route.

Finger endurance test #2: Determine the longest you can hang from a fingerboard 20mm (or similar) edge using the half-crimp or open-hand grip. Your results will likely be somewhere between twenty seconds and two minutes and, therefore, this testing protocol provides a good measure of your anaerobic lactic energy system. Warm up extensively—be sure to

exercises such as fingerboard and campus training will carry over to climbing better than nonspecific exercises such as finger rolls, squeeze devices, and the like.

4. The exercise must focus on a specific grip position for an entire set. In climbing, the rock dictates a random use of varying grip positions. Since strength is grip-specific, such cycling of grip positions allows you to climb longer than if you used the same grip repeatedly. That's great if you are climbing for performance, but it's ineffective for building maximum grip strength. (This

helps explain why a full season of climbing builds local forearm endurance but often leaves you with about the same maximum grip strength as before.) Effective grip-strength training must target a specific grip position to near failure—I recommend training the half crimp (or open crimp), open hand, wide pinch, and the three two-finger pocket grips.

Keep these four requirements in mind as you execute the maximum-strength finger exercises detailed below, especially the second requirement: Be sure to scale the exercise resistance to be near maximal (near failure in under twelve seconds).

get through the flash pump phase—before taking this test. Ask a partner to give you verbal encouragement, and do feel free to pull up with your arms, if needed, to obtain a true maximum-length hang. Do two tests with at least a fifteen-minute rest in between.

The Zlagboard finger training system (see photo) is an excellent testing—and training—platform for coaches and serious climbers. This patented system incorporates a nifty switching mechanism that marries your smartphone to the Zlagboard, thus providing precise hands-free operation that directs and records hang-time measurements, archives past hang tests, provides professional training programs, and allows you to compete with climbers locally and globally.

Based on recent Zlagboard Contest results, the median hang-time is about forty-five seconds—a respectable time that is commonly achieved by accomplished climbers who climb at a moderately high level. Using the Zlagboard 20mm "contest holds," hang times between sixty to ninety seconds are common among advanced climbers, typically capable of climbing 5.13. Hangs of more than ninety seconds indicate elite-level strength-endurance (often 5.14 climbers). And if you can hang more than two minutes . . . then you're likely a famous climber already! The current world-record hang (as of March 2016) is an



incredible 2:44 by Spanish rock star (and two-time world champion) Ramon Julian Puigblanque. Learn more at: www.ZlagboardUSA.com.



BOULDERING

Bouldering is the most straightforward way to train grip strength. Without the constraints of a rope and gear, bouldering allows you to focus on climbing the hardest moves possible. Inherent to hard bouldering, however, are some limiting factors that diminish the potential to build maximum grip strength. Consider that technical difficulties may prevent you from climbing to near the point of muscular failure. Furthermore, the rock or plastic

dictates the use of many different grip positions, thus making it difficult to isolate a single grip position and work it to failure.

Despite these limitations, bouldering should be a staple of your training program. It will build some functional strength and, at the same time, develop your mental and technical skills. Consequently it's a good training strategy to couple bouldering with one of the other finger-strength-training exercises described in this section. Use the following training strategy to best stimulate gains in finger strength via bouldering.

Select a short boulder problem that appears to be strenuous, but not technically difficult. Favor overhanging problems, which will place more weight on your hands and maximize the training effect. When bouldering indoors, try to locate (or consider setting) problems that isolate a specific grip position; for example, a problem that possesses a lot of crimp holds will be best for training crimp strength. The ideal boulder will be climbable in under fifteen seconds, since the goal is to target the anaerobic alactic energy system—if you're getting seriously pumped, then the problem is too long (bringing the anaerobic lactic system deeply into action). Attempt to climb the problem three times with sufficient rest (two to three minutes) between each ascent to allow a good effort. Doing a problem three times makes one "set."

Move on to another strenuous-looking problem that appears to target a different grip position, such as wide pinch, two-finger pockets, open hand, and such. Ascend this problem three times (one set), with adequate rest (two to three minutes) between attempts. Continue bouldering in this way for thirty to sixty minutes (three to ten total sets), and then finish your finger training with one of the isolation exercises described later in this section.



HYPERGRAVITY BOULDERING

Advanced climbers with several years of bouldering under their belt eventually reach a point where they no longer achieve significant gains in finger strength despite regular, hard bouldering. Fortunately, hypergravity bouldering, weighted fingerboard hangs, and the HIT System workout (all detailed below) are potent training methods that will yield further gains in high-end finger strength. To do this, you'll need to invest in a five- or ten-pound weight belt (twenty pounds for an advanced climber) or fill a fanny pack with five to ten pounds of scuba divers' weights. Here is the best protocol for engaging in hypergravity bouldering—this is an indoor training strategy only!

Complete a general and specific warm-up—that is, work through various warm-up and mobility exercises, and then move on to some moderate climbing and bouldering lasting at least thirty minutes. Now clip on your weight belt and predetermine a target number of hypergravity bouldering sets (repeating a problem three times makes one "set") that you will perform. As a guideline, limit yourself to two sets on your initial session, then build to five to seven sets as you gain confidence and strength.

Select short, overhanging boulder problems that possess small- to medium-size holds but no tiny and tweeky features, and no dynamic or highly technical moves. Since you are climbing with a weight belt, favor problems that are a couple of grades or more below your limit—you should be able to ascend the



Hypergravity bouldering with weight belt.

Grip Positions

Although rock dictates a wide range of gripping positions, we can best train maximum grip strength by addressing the six primary grip types. Let's take a quick look at each grip position and the training considerations.

Full Crimp

The full crimp is favored by many climbers because it provides what feels like the most secure lock onto small handholds thanks to its hallmark thumb lock over the top of the tip of the index finger.

The full-crimp grip excels on small, square-cut edges, shallow flakes, and any hold that possesses a small incut edge. Unfortunately, the full-crimp grip is the most stressful on the tendons and joints due to the hyperextension of the first (distal) joint of each finger and the sharp flexion of the second finger joint. Chronic use of this grip can lead to various tendon injuries in the fingers and elbows, especially among individuals who frequently chicken wing their elbows. It's best not to specifically train in this grip position, but instead focus on the half-crimp grip (sans thumb lock) since it's a bit less stressful on the joints and tendons, and the training gains will largely transfer to the full-crimp position.



Half Crimp

The half-crimp grip is a slight variation of the full crimp in which you do not thumb-lock over the index finger. This reduces tendon stress and softens the aggressive angles on the fingers' first and second joints, but at the expense of a slightly less secure feel. Fortunately, targeted training



of the half-crimp grip will yield large increases in strength that will make a huge difference on the rock, especially when crimping on small edges, incut pockets, and narrow pinches.

Open Crimp

The open crimp is a hybrid grip characterized by the two middle fingers clinging in a half-crimp fashion, while the shorter index and pinky fingers assume the extended open-hand grip. The open crimp is often used in fingerboard training and in climbing crimpy endurance-oriented routes.



Open Hand

The open-hand grip has distinct advantages over the crimp grip. First, the open-hand position is much less stressful on the finger joints and tendons—one study (Vigouroux 2006) revealed that the force on the A2 pulley is thirty-six times lower in the open-hand grip than in the crimp grip. Furthermore, despite its less secure feel, the open-hand position can be trained to become your strongest grip position on all but the smallest crimp and incut holds (which require a crimp grip). This grip is most effective on deep, rounded or sloping holds, and particularly when pulling on pockets. If



you're unfamiliar with the open-hand grip, it will feel quite awkward and difficult at first. But rest assured that your open-hand grip strength will improve quickly with targeted training.

Pocket Grip

This is not really a unique grip, but it's worth including here since the pocket grip deserves some dedicated training. While shallow or incut pockets will require a half-crimp-like grip, most pockets (deeper and rounded) are best engaged with fingers extended in the open-hand position. Train the open-hand pocket grip with all three pairs (or "teams") of fingers. Advanced climbers (solid 5.13 and seven-plus years' experience) can ease into training one-finger pockets with the middle finger.



Pinch Grip

The pinch grip is vital for latching on to protruding holds such as pebbles, tuffas, aretes, and opposing edges. Due to the protruding nature of indoor climbing holds, use of the pinch grip is far more common indoors than on natural rock faces. Fortunately, our hands are designed to excel at pinching thanks to strong thumb muscles that help anchor this grip. An important distinction is the difference between narrow and wide pinch grips. Pinches of less than about three inches in width place the fingertips in a half-crimp-like position (hyper-extended distal interphalangeal [DIP] joint), whereas wider pinches extend the fingers into an open-hand position with an over-extended wrist. While it would be smart to train both narrow and wide pinches, it's the wide pinch grip that should be your primary focus.



route in less than fifteen seconds (anaerobic alactic system). Climb the problem three times with a rest of at least two to three minutes between ascents—these three ascents (or attempts) make one set.

Move on to another strenuous problem that appears to target a different grip position. Consider taking the time to set "theme problems" that possess only holds of a certain shape and size—this is the best way to target and train a weak grip position. Do one set of three ascents of each problem, always resting for at least two minutes between each ascent.



FINGERBOARD "MINIMUM EDGE" HANGS

The following training protocol, loosely based on research by Eva Lopez-Rivera (2012), is what I

recommended as an entry-level fingerboard program. A few multiweek cycles will bring noticeable gains in finger strength for intermediate climbers, as well as serve as preparation for delving into weighted fingerboard hangs (described next).

You'll need a fingerboard with many different-size edges and pockets, and then you'll need to experiment a little to identify what features you can hang on for just fifteen seconds or so. Actual training hangs should terminate a few seconds before failure of the grip, so I recommend making each hang exactly twelve seconds in length (the terminus of the anaerobic alactic energy pathway). Initial fingerboard workouts should focus on training with either the half-crimp or open-crimp grips, although you can eventually add in some open-hand pocket



Fingerboard train only with the open-hand, open-crimp, and half-crimp grips. Eva Lopez-Rivera's Transgression board is pictured.

and pinch grip training, if your fingerboard allows. Never fingerboard train with the full-crimp (with thumb lock) grip!

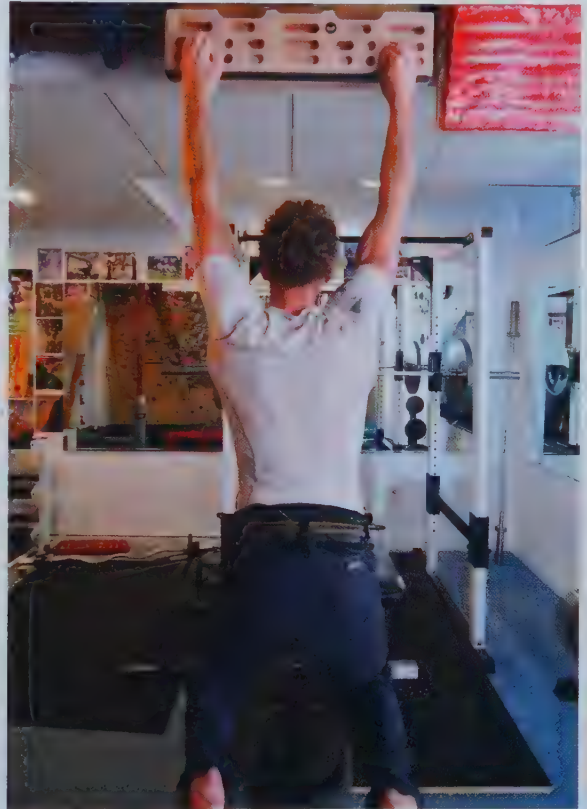
The training protocol is simple: Do a series of five, twelve-second hangs with exactly three minutes rest in between. Each hang should be near-maximal (rating of perceived exertion of 9 to 9.5 out of 10), but not quite take you to failure—as mentioned above, use a feature that you can barely hold for fifteen seconds with maximum effort. Begin by doing just two sets, but more advanced climbers can do two or three additional sets to isolate other grip positions. Always rest five minutes between sets (see table 8.2).

By design, this training protocol will produce little or no muscle pump, as it primarily targets the anaerobic alactic energy system.



FINGERBOARD “MAXIMUM WEIGHT” HANGS

A period of dedicated “minimum edge” fingerboard training will eventually lead you to the point of needing to train on tiny, painful edges to keep with the protocol detailed above. Enter the “maximum weight” protocol, which involves using larger, more



The author maximum weight training with eighty pounds hanging from the belay loop of his harness.

Table 8.2 “Minimum Edge” Fingerboard Protocol

Duration of Each Hang	Rest Between Hangs	Hangs per Set	Intensity (scale of 1 to 10)	Number of Sets	Rest Between Sets
12 seconds	3 minutes	5	9–9.5	2–5	5 minutes

Note: Select holds that you can barely hold for 15 seconds, although all hangs must terminate at 12 seconds.

Table 8.3 Maximum Weight “10-Second” Protocol

Duration of Each Hang	Rest Between Hangs	Hangs per Set	Intensity (scale of 1 to 10)	Number of Sets (aggregate of all grips trained)	Rest Between Sets
10 seconds	3 minutes	5	9–9.5	2–5	5 minutes

Note: Your training weight should be heavy enough to cause failure in 13 seconds, although all hangs must terminate at 10 seconds.

Table 8.4 Maximum Weight “7-53” Protocol

Duration of Each Hang	Rest Between Hangs	Hangs per Set	Intensity (scale of 1 to 10)	Number of Sets (aggregate of all grips trained)	Rest Between Sets
7 seconds	53 seconds	3	9–9.5	2–5	5 minutes

Note: Your training weight should be heavy enough to cause failure in 10 seconds, although all hangs must terminate at 7 seconds.

comfortable holds, but with added weight to create sufficiently high intensity for maximum-strength adaptations. The ideal size edge for weighted fingerboard hangs is between 14mm and 20mm ($\frac{5}{8}$ to $\frac{3}{4}$ -inch), or a little less than one finger-pad depth—this moderate depth lessens skin pain, reduces strain on the distal (DIP) joint, and has been proven effective to develop maximum-strength gains that will carry over to different-size edges on the rock.

Tables 8.3 and 8.4 present two maximum weight training protocols that really work. The first is adapted from Eva Lopez-Rivera’s research, whereas the second is my own advanced protocol to train both maximum strength and aerobic power (increased rate of CP resynthesis between hangs). Do two to five sets of one or the other—not both!—focusing mainly on the half-crimp or open-crimp grips, although advanced climbers may want to dedicate one or two sets to pocket or wide pinch grips. The amount of weight added will be significant (generally between twenty-five and one

hundred pounds), so you’ll need to invest in a large weight vest, several weight belts, or hang a combination of free weight plates from the belay loop of your harness. The latter is the best solution since you can easily unclip the weights to de-load your body between hangs and sets.

Record your training details in a notebook, including hold size, weight used, and the number of hangs and sets. In the months and years to come, you’ll undoubtedly document some significant gains in finger strength!



HYPERGRAVITY ISOLATION TRAINING— MAXIMUM-STRENGTH PROTOCOL

Hypergravity Isolation Training (HIT) is a cross between hypergravity bouldering and weighted fingerboard hangs. The unique HIT System provides a platform on which you can target a specific grip position for a single set of wall climbing. Since HIT System training involves actual climbing (unlike fingerboard training), it calls into action the entire

Overview of Fingerboard Training

Since its advent in the mid-1980s, the fingerboard has become the most used type of training equipment among climbers—and for good reason: The straight-armed, weighted hang is the single most effective isolation exercise a climber can do. What's more, the fingerboard is economical, and it can be mounted in just about any apartment or home.

The obvious strengths of fingerboard training are its ease of access and the ability to isolate a wide variety of grip positions. While not appropriate for beginners, experienced climbers can progressively add weight to their body to train maximum grip strength with a series of brief, high-intensity hangs. The strategy for training strength-endurance is to use lighter loads and a higher number of hangs, but with less rest in between.

Being able to vary the training load is an important aspect of effective fingerboard training. While you can indeed adjust intensity up and down by using smaller and bigger holds, respectively, it's also important to be able to adjust resistance while training on a specific hold such as the common 20mm (3/4-inch) edge. To increase resistance, simply wear a weight vest or hang free weights from the belay loop of your climbing harness. For resistances less than body weight, you can employ a pulley system with counterweights—especially useful for training one-arm hangs and learning one-arm pull-ups.

Always train with good hangboard technique: Maintain moderate tension throughout your shoulders and upper torso by engaging your scapular stabilizers (think about pushing your chest out); relax from the hips downward and avoid lifting your knees excessively; and most important, do not relax your shoulders and allow them to elevate into an extreme shrug position near your ears. Also, it's vital to do preparatory and concurrent training of the scapular stabilizers and rotator cuff muscles—select from exercises detailed in chapter 6.



Two-finger pocket pull-up training on a NICROS Nexgen fingerboard.

A word of caution: Misuse of the fingerboard has contributed to tendon injuries in countless climbers. Fingerboard training should be limited to just two days per week and, ideally, as a supplement to climbing rather than a replacement for actual climbing. A gradual warm-up is essential beforehand, including a general activity to elevate heart rate, followed by various mobility exercises and some self-massage as outlined in chapter 6. Complete your warm-up with some pull-ups on large holds. It's also a good idea to conclude your fingerboard training with a few sets of antagonist training of the wrist stabilizers.

chain of climbing muscles, from the finger flexors to the arms to the many core muscles of the torso. This novel exercise is unquestionably one of the best pathways to becoming a stronger climber.

It's ideal to use the HIT workout as part of a training cycle or an off-season program; however, you can also use HIT once weekly during your climbing on-season. As part of an off-season (winter) program, you might do four to six HIT workouts over the course of a two- or three-week maximum-strength mesocycle, and then switch to another training modality (aerobic or anaerobic endurance) for a few weeks. Given its stressful nature, it's best to cycle on and off this exercise every couple of weeks. Learn more about the HIT System in the sidebar on page 191.

The maximum-strength HIT System protocol (Table 8.5) trains six basic grip positions in sets lasting a maximum of fifteen seconds—no set should go longer, otherwise the training focus becomes anaerobic lactic. The grips trained are half crimp, pinch, open hand, and the three “teams” of two-finger grips. There will be enough near transfer from these fundamental positions to make your fingers stronger in any configuration on the rock. Perform one or two sets for each grip position—those new to the HIT workout should begin with one set—starting with the most difficult grip position for you. Most people work through the grips in this order: pinch, two-finger “third team” or outside pair (pinky and ring finger), two-finger “second team” or inside pair (index and middle finger), two-finger “first team” or middle pair (middle and ring finger), half crimp, and open hand. The entire HIT workout is done “open feet,” meaning that you can place your feet on any holds on the wall.

Here's how you would perform a HIT set on the pinch holds: Sitting below the first HIT Strip, begin by gripping the first set of right- and left-hand pinch holds, then pull up and grab the next higher left-hand pinch hold. Continue climbing with the next higher right-hand pinch hold and the

next higher left-hand pinch hold until both hands are on the top two pinch holds. Begin descending immediately, alternating left and right pinch holds back down until you are holding the bottom two pinch holds—this will take you nearly fifteen seconds, so end the set now. Upon stepping off the wall, use a stopwatch or phone app to time a rest of exactly three minutes before beginning the next set. Meanwhile, record the total number of “reps” (hand movements) and weight used in your training notebook. Use weight belts to add weight, since having the extra weight near your center of mass will have the least impact on your climbing technique; above about thirty pounds you'll need to use a weight vest.



Training on the HIT System pinch holds with a twenty-pound weight belt.

Table 8.5 HIT System—Maximum Strength Protocol

Grip Position	Weight Used HIT “Novice”	Weight Used HIT “Advanced”	Set Duration (reps/set)	Number of Sets	Rest Between Sets
Pinch	None	20 lbs.	<15 seconds (8–15 reps)	1–3	2–3 minutes
2-Finger “3rd Team”	None	20 lbs.	<15 seconds (8–15 reps)	1–3	2–3 minutes
2-Finger “2nd Team”	6 lbs.	30 lbs.	<15 seconds (8–15 reps)	1–3	2–3 minutes
2-Finger “1st Team”	8 lbs.	40 lbs.	<15 seconds (8–15 reps)	1–3	2–3 minutes
Half Crimp	10 lbs.	40 lbs.	<15 seconds (8–15 reps)	1–3	2–3 minutes
Open Hand	10 lbs.	40 lbs.	<15 seconds (8–15 reps)	1–3	2–3 minutes

* Weights are approximations for a 160-pound climber. Use similar percentages of your body weight.

After your three-minute rest, proceed immediately with a second set of pinch grip (advanced) or begin training the next grip position (two-finger “third team”) by utilizing the identical two-finger pocket holds on each HIT Strip. Once again climb up and down the HIT Strips one time using only this grip position. The commonly weak two-finger third team and pinch grips may require little or no additional weight, but if doing one full up-and-down lap is easy, then you will need to add weight next time. Ideally, you want to add enough weight to make ten consecutive hand moves difficult (effort of 9 out of 10). Rest for two to three minutes before performing the second set with this grip (advanced) or moving onto the next grip position. Continue in this manner until you’ve trained all six grip positions—a hardy workout, for sure!

Contact Strength Exercises

Contact strength is your ability to quickly latch onto a hold. This trait is directly related to the speed at which you can recruit the forearm muscle’s motor units and summon peak strength—a capacity called rate of force development (page 109). While the maximum-strength exercises described earlier will yield some improvement in contact strength, reactive training exercises that emphasize speed and shock loading is the optimal method for increasing contact strength. Since fast, dynamic movements are fundamental to effective reactive training, the resistance used (training load) must be significantly less than in the maximum-strength exercises. For many climbers the resistance will need to be less than body weight to allow for the rapid movement and turnover (change in direction) that’s essential for effective reactive training.

Why Do Strong Fingers Matter?

Let me count the ways...

1. Stronger fingers can grip smaller holds.

Duh! Seriously, though, all the climbing in the world won't do much to improve your ability to grip small edges and pockets. Improvement at gripping smaller and smaller holds—essential for climbing harder—depends on building a higher level of absolute (limit) grip strength. Fingerboard training (with the right training protocol) is unquestionably the best method of training maximum grip strength.

2. Stronger fingers can endure longer when climbing on submaximal holds.

As explained earlier, when your fingers are gripping at just 20 percent of maximum voluntary contraction (MVC), blood flow through your forearm flexor muscles begins to slow ... and at a contractile force of about 50 percent of MVC, the blood flow is occluded completely. When blood flow is reduced or stopped completely, power output declines precipitously after about ten seconds and, depending on the difficulty of the sustained grip (percent MVC), failure will follow in a matter of seconds to, at best, a minute or two if the finger flexors are contracting at a force far less than 50 percent MVC.

Anyway, lack of blood flow and oxygen supply is the crux of the matter, since oxygen is required for CP resynthesis and blood flow is needed to remove metabolic byproducts of anaerobic lactic

energy production. Thus, when resistance climbing through a long sequence of relatively difficult, yet submaximal holds, the greater the perfusion (delivery of blood to the capillary bed) the longer you will endure. Given the above understanding of blood flow occlusion as a function of percentage of maximum voluntary contraction (MVC), it's now easy to understand that in gaining a higher level of maximum grip strength (a greater MVC) you will be able to support your body weight (or whatever part your feet aren't supporting) with a smaller percentage of MVC. This is a powerful concept and a critical guiding principle for effective training!

Summarizing: Increasing the MVC of the finger flexor muscles enables contraction at a lower percentage of maximum (compared with the weaker fingers of the "old you") when climbing on similar submaximal terrain, thus allowing for increased blood flow (and use of the aerobic energy system) and improved endurance.

3. Stronger fingers can rest and recover on smaller holds. I'm sure you've seen videos of rock stars, such as Chris Sharma and Adam Ondra, pausing in the middle of some long, steep, and ridiculously hard climb to shake out and somehow rest on a small "finger bucket" that would in no way be a rest hold for the mass of climbers. Yet somehow

Reactive training is appropriate only for intermediate and advanced climbers with no recent history of finger, elbow, or shoulder injury. Begin with the tamer feet-on exercises (less than body weight) and progress gradually to the feet-off exercises over the course of months or years. A few sets of reactive training will impart all the stimuli necessary for favorable adaptations—resist the urge to perform additional sets. Remind

yourself that such neural training does not produce extreme fatigue, so if you train to the point of high fatigue, you are doing far too much and tempting injury.

Following are five reactive training exercises listed in order from least stressful (lowest force load and injury risk) to most severe (highest force and injury risk). Controlled one-arm lunging is the ideal icebreaker exercise for climbers wishing to

these guys can stop and catch their breath and recover enough power reserve to push on and finish the climb. What is their secret to resting on holds so small they'd pump out an average climber? Strong fingers, of course!

As explained in #2, blood flow is the key to resynthesizing CP (essential for brief, high power movements) and removing fatiguing metabolic byproducts of the anaerobic lactic energy system (the predominant energy system when climbing hard, sustained sequences lasting between twelve seconds and a minute or two). For a rest hold to be good enough to hang onto for a significant rest (thirty to sixty seconds or more), you must be able to grip the hold using less than 50 percent of your MVC—only this way will there be any rejuvenating blood flow. Therefore, a climber with “average” grip strength may require a four-inch deep hold to recover (the smallest size hold he can grip for an extended time at <50 percent MVC) whereas the 5.15 climber can likely rest on a one-inch hold (with feet on the rock, of course) thanks to their incredible grip strength that allows them to grip this small hold with less than 50 percent of MVC.

4. Stronger fingers have more stamina.

Occasionally you hear a mind-boggling media report of some pro climber on-sighting or redpointing multiple 5.14 routes in a day and then doing it again the next day . . . and the next! How is that even possible?

add some reactive training to their routines, as each one-arm catch involves a slight shock loading of the forearm muscles. Contact strength gains achieved through this form of feet-on “reactive light” training are limited, however—beyond a certain point you will need to graduate to feet-off campus training, which produces a much higher shock load, to stimulate further gains.

Let me tell you a story about a climber named Alex who does this with regularity. I've climbed with Alex a few times in recent years, and in October 2015 I was with him at the New River Gorge when, in the matter of a few hours, he warmed up on a couple 5.13+ routes and then proceeded to do a first ascent of a 5.14b and a second-go second ascent of another 5.14b. Sounds like a fatiguing day of climbing, right? Not for Alex. In fact, he climbed other routes at the 5.14 grade both the day before and the day after!

How is this possible? Before I embellish further on the importance of getting stronger, I must point out that Alex possesses world-class technique and a very strong mind. That said, the master key is that Alex is really strong! (See the front cover photo of this book.) We have yet to see what Alex's true climbing limit is (but it will be fun to watch in the coming years), but I can tell you that climbing 5.13+/5.14- routes are nowhere near his limit. So for Alex to climb at this level on successive days is similar to a 5.12b climber doing 5.11 routes on successive days or a 5.14a climber sending multiple 5.12+/5.13- routes.

The bottom line: By increasing your limit strength and advancing your high-end climbing ability, the submaximal level at which you can climb with relative ease and aplomb—and in high volume—is elevated. Therefore, by making a long-term training commitment to getting stronger (and improving your mental and technical game), you may someday find your current maximum climbing level to be “moderate” and achievable in volume.



ONE-ARM LUNGING (FEET-ON)

One-arm lunging with feet fixed on good holds (or the floor) is the easiest reactive training exercise. You can do this exercise using the bottom two large holds on a campus board or two juggy holds at the base of a slightly overhanging section of climbing wall. Ideally you can set a few modular holds specifically for performing this exercise. Set two footholds



1. Grip the higher handhold.



2. Drop down to the lower.



3. Lunge back to the higher.

about a foot off the ground, and then set two large, rounded holds that fit your hand nicely (open-hand grip), one in front of your face and the other about ten to eighteen inches above that. If training on a campus board, you'll need to lunge between the two lowest large holds on the board with your feet on the floor below and slightly behind the board.

Either way, balance your weight evenly on both feet and then grip the higher of the two jug holds with both hands. Now let go with one hand and hold the arm by your side or behind your back. Begin with an inward pull with the engaged hand, and at the deadpoint (moment of quasi-weightlessness on that hand) let go and drop the hand down to catch the lower jug hold with a partially bent arm and engaged rotator cuff and scapula (think about keeping your chest out). Absorb the downward energy and immediately recoil upward

to catch back on to the starting handhold. Continue with brisk up-and-down lunging for six to twelve total hand movements—total duration of exercise is ten seconds or less.

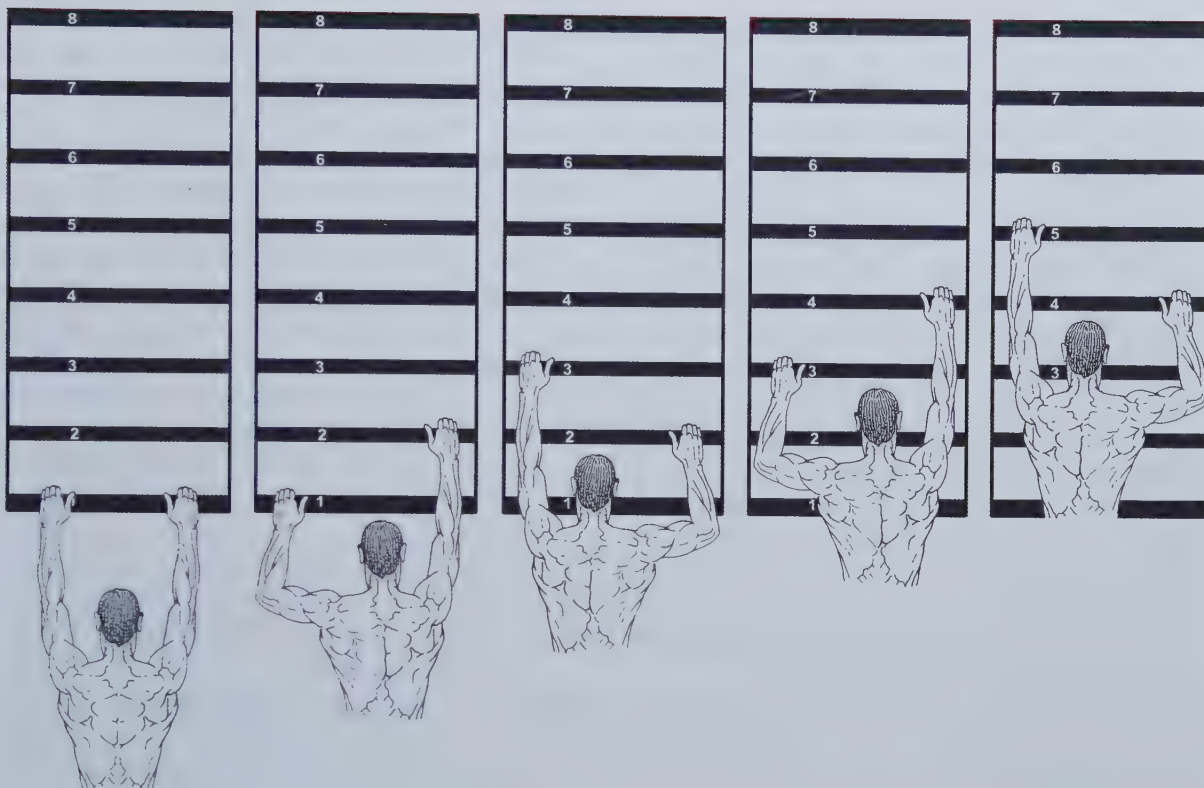
After a brief rest, step back up on the wall and perform an identical set of one-arm lunges with the other hand. Perform two or three total sets with each hand.



CAMPUS LADDERING

As the name implies, this exercise involves climbing in a hand-over-hand, ladderlike motion up the campus board with no aid from the feet. Unlike double dynos (described later), this laddering exercise uses controlled dynamic movements that are less likely to result in injury. Consequently, this is a better staple exercise for regular use, and it is the

Figure 8.1 Campus Laddering



perfect icebreaker exercise for climbers new to campus training. An important distinction when ladder training: Using smaller holds and shorter reaches trains contact strength whereas using larger holds and longer reaches trains pulling power and lock-off strength (the subject of chapter 9). The focus here is on contact strength, so you should use the smallest size holds that you can ladder without skipping rungs.

Hang with nearly straight arms from the bottom rung of the campus board. Your hands should be about shoulder width or slightly less apart. Striving for brisk, fluid motion, climb hand-over-hand up the campus board using alternating rungs for your

left and right hands. Your goal is to ascend the board as fast as possible and, upon reaching the top, jump off. Most campus boards feature at least eight rungs, so your training goal is to make eight hand moves as quickly as possible. Increase difficulty by using smaller rungs or adding five to ten pounds of weight around your waist rather than skipping rungs.

Think of this exercise as an alactic sprint—laddering eight rungs should take just three to six seconds. Initially, rest two to three minutes between sets; however, you can train aerobic power by decreasing the rest interval to just thirty to sixty seconds between sets. Begin by doing just three sets, but build toward a maximum of ten sets.



CAMPUS “BUMPS”

As the name implies, this exercise involves bumping one hand up successive rungs while the other hand remains fixed on the bottom rung. Like most campus board exercises, you’ll be using momentum to assist with the upward hand bumps—still, each rung “catch” will elicit a high rate of force development in the finger flexors, which is of course the training goal here. Begin by hanging from the bottom rung with shoulders engaged (not shrugging) and arms just slightly bent, and then proceed to pull upward and “bump, bump, bump, bump” the active hand as fast as possible while the other arm pulls down into a lower and lower lock-off position.

Given a campus board with normal-size rungs, you’ll likely be bumping the active hand just three

or four times before you jump off and call it a set. Importantly, your highest bump hold must not be with a straight arm and shrugged shoulder—this is a critical distinction, as many shoulder injuries occur when individuals repeatedly campus with fully extended arms and elevated scapula. As with the previous exercise, increase difficulty by using smaller holds or adding a small amount of weight rather than skipping rungs; make the exercise easier by placing your feet on holds fixed below the campus board. Initially do just two or three sets on each side, although in the long term, advanced climbers can progress up to ten sets each side. Resting two minutes between sets is usually sufficient; however, advanced climbers can reduce rests to just thirty seconds to train aerobic power.

Figure 8.2 Campus Bumping



Overview of Campus Board Training

Developed in the late 1980s by Wolfgang Güllich at the Campus Center in Nürnberg, Germany, campus training is the gold standard for training upper-body power and contact strength for climbing. The campus board is a unique training platform that allows for various reactive training and quasi-plyometric exercises that train recruitment and rate of force development in the finger and arm flexors. This type of dynamic training is a real boon for high-end sport climbers and boulderers.

While feet-off “campusing” is the most common method of training among elite climbers, feet-on campus training with the toes resting on small holds or wooden strips is the preferred method for more intermediate climbers wanting to dabble on the campus board. An ideal campus board rig will include at least three different-size strips (roughly 20mm, 40mm, and 60mm “jug” or open-hand sloper) and foot strips at a few different heights under the board.

Sadly, the campus board is widely misused, both by individuals not strong or experienced enough for the high dynamic loads and by some elites who overutilize this training platform. No one with less than two years climbing experience should use a campus board—their tendon strength is unlikely to be ready to carry the load. Even elite climbers should limit campus training to less than 10 percent of total training time. Finally, anyone wanting to campus train should first engage in several months of scapular stabilizer training (see chapter 6) and continue with this training indefinitely. Here are several more guidelines for safe and effective use of the campus board.

Engage in campus training only if you are an advanced-intermediate (leading 5.11 or bouldering V5) to elite climber with no recent history of finger or arm injury. Favor feet-on exercises and feet-off laddering and bumps. The most stressful double dynos exercise must be reserved for only highly conditioned, elite climbers (5.13/V8 climbers).

Warm up thoroughly. Spend at least half an hour performing various warm-up activities and bouldering to fully warm the finger, arm, shoulder, and back muscles.

Use only the open-hand or open-crimp grip when doing feet-off campusing.

Never engage a rung with a completely straight arm and shrugged shoulder—this will wreck your shoulders in short order! Upward grabs should always be done with a slight elbow bend and a well-secured scapula (think about keeping your chest up/out rather than caved in).

Emphasize quality over quantity—it's better to do a few perfectly executed explosive sets rather than many reckless sets with poor technique (straight arms and shrugged shoulders).

Do not campus train while in a state of high fatigue or if you have any doubts about the health of your fingers, arms, or shoulders.

Immediately terminate your campus training at the first sensation of pain in your fingers, elbows, or shoulders (common among people who campus with poor technique and weak scapular stabilizers).

Campus no more than twice per week, and cycle on and off campusing every couple of weeks.

Youth climbers should abstain from campus training, especially during peak growth velocity (ages thirteen to sixteen) when growth plates are most easily injured.

The famous Café Kraft campus board in Nürnberg, Germany, the birthplace of campus training!

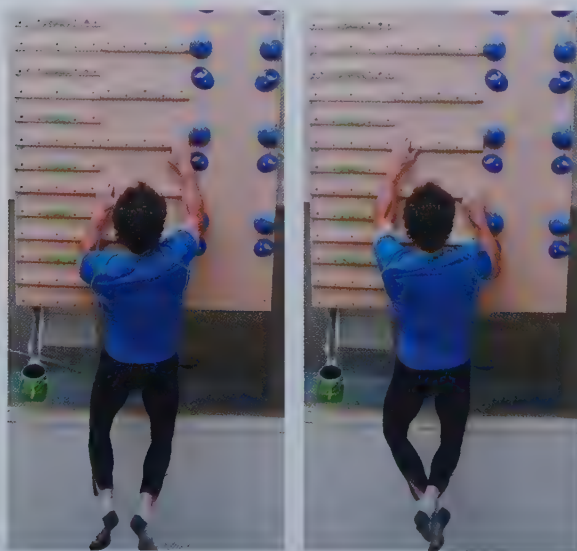




CAMPUS “SWITCH HANDS”

Whereas the previous campus training exercises all involved advancing one hand while the other hand remained engaged on a rung for support, the Switch Hands exercise escalates to moving both hands simultaneously, albeit for just a small distance of about ten inches. Still, this dynamic movement demands more rapid rate of force development in the finger flexors to successfully re-grasp the rungs after the double release move. The Switch Hands exercise, therefore, is a good segue toward eventually performing the most difficult double dynos exercise (below).

Do Switch Hands using positive medium to large-size rungs (not slopers) and an open-hand grip. Begin with your hands on successive rungs (either the first and second, or the second and third), and then tighten down your scapula (think chest out) and bend your arms slightly to gain a good starting position before lifting your feet off the floor. Begin the exercise with a short but sharp pull—only a few inches—to create upward momentum and then, at the deadpoint, quickly switch hands to the opposite rungs. You’ll latch



Campus Switches

back on to the rungs as your center of mass begins to descend, so you’ll need to momentarily absorb this energy with your pulling muscles before initiating the next upward pull and hand switch. Continue switching hands as fast as possible, for up to ten or twelve total hand switches.

There’s a little bit of timing involved in doing this correctly, but if you’re strong enough to rightfully employ this exercise you’ll quickly acquire the skill and be able to do about ten to twelve successive switches in only about five seconds. This is a bang-bang exercise that, if done correctly, involves very little upward and downward movement of the torso. The exercise itself is pure anaerobic (alactic) power; however, it’s also excellent for developing aerobic power (for faster resynthesis of CP) if you do several sets in a row with only short rests. Begin with just two sets per session with a two-minute rest in between, but gradually build toward six sets with just thirty to sixty seconds between sets (advanced).



CAMPUS DOUBLE DYNOS

The double dyno exercise is the pinnacle of campus training with its double-handed, fully airborne flight between rungs. Done on smaller rungs and without skipping (rungs), this exercise focuses more on finger flexor recruitment, whereas using large rungs and greater flight distance (skipping rungs) equally trains finger and pull-muscle power. The focus now is on developing finger contact strength via more synchronous motor unit recruitment and neural disinhibition, so the protocol here is smaller rungs and no rung skipping.

Every climber must acknowledge, however, the extremely stressful and potentially injurious nature of this exercise. Are your fingers tendons, rotator cuff, and scapular stabilizers ready for this level of stress? It’s my belief that the answer is “no” for over 95 percent of climbers. I discourage use of this exercise if you’ve been climbing less than four

Complex Training

Complex training is a cutting-edge training method used by elite athletes in many sports, including most power-oriented Olympic events. Applied to climbing, the complex training protocol described below is one of the most advanced strength-training concepts available. Since introducing complex training in the first edition of *Training for Climbing* in 2002, I have heard from countless climbers around the world who have leveraged this technique to increase their grip strength and upper-body power. You can, too, as long as you are an intermediate-advanced climber (solid at 5.11 and V5 or harder) with no recent history of finger, elbow, or shoulder injury.

Complex training involves a coupling of a maximum-strength exercise with a reactive power-oriented exercise. Research has shown that performing these two very different exercises back-to-back—and in the order of strength first, power second—produces gains in strength and power beyond those achieved by performing either exercise alone. While no studies have been done with climbers, there is compelling research in the use of complex training to increase vertical jump that shows phenomenal gains in absolute ability (Adams 1992). In this study six weeks of strength training produced a 3.3-centimeter increase in the vertical jump, compared with a 3.8-centimeter increase after six weeks of reactive (plyometric) training. The group performing complex training (strength and reactive) for six weeks experienced an incredible 10.7-centimeter increase in jumping ability!

years, don't climb 5.13 or V9, can't do at least five hypergravity pull-ups with one-third of your body weight added, and can't easily ladder up and down the campus board in a 1-3-5-7-7-5-3-1 sequence.

Begin by hanging from one of the middle rungs, with slightly flexed arms and engaged

Incorporating complex training into your program can be done a number of ways—the key requirement is a coupling of a maximum-strength exercise and a power exercise with minimal rest in between. Here are a few common training complexes:

- Do a fingery near-maximal boulder problem followed immediately by one set each hand of one-arm lunging (feet-on) on a campus board or climbing wall.
- Do a hypergravity boulder problem (with, say, a twenty-pound weight belt on) coupled with a set of Campus Laddering (sans weight belt).
- Do a max-strength set on the HIT System (with weight belt) followed immediately by a set of Campus Switch Hands (no added weight).
- Do a max-strength set of fingerboard hangs (with weight) followed by a set of campus board double dynos. This training complex is one of the most powerful for elite climbers, and it may represent the single best training protocol for pursuing absolute genetic potential for contact strength. Elites can build to doing up to eight coupled sets (complexes) with three to five minutes rest between sets.

Cautionary note: Complex training is, obviously, an advanced technique that produces both high passive and dynamic stress on the finger tendons and joints, the elbows, and the shoulders. This protocol should be used only by well-trained climbers with no recent history of injury. Furthermore, limit use of training complexes to once every three or four days and primarily during the maximum-strength and power mesoscale.

shoulders. Simultaneously let go with both hands and drop to catch the next lower rung with flexed arms and engaged shoulders. After a brief amortization phase of energy absorption in the pull muscles, explode upward with both hands to catch the starting rung. This is one full repetition, but



Campus Double Dynos

don't stop! Without hesitation, drop down to again catch the next lower rung, and explode back up to the starting rung. Continue for three to five repetitions (six to ten total hand moves), and then dismount. The turnaround time must be as fast as possible—ideally less than one-half second—and the total exercise time is less than ten seconds. This exercise is almost fully alactic powered, so you should get hardly a hint of pump in doing it. Do two to five sets with two to three minutes rest in between each set.

Warning: End the exercise prematurely rather than risk a failed downward catch (but have a bouldering pad in place just in case). Terminate your campus training at the first sign of pain in your fingers, arms, or shoulders. Remember rule number one: *Never get injured training!*

Strength-Endurance Exercises

As detailed in chapter 5, your ability to climb many hard moves in a row is largely a matter of your anaerobic capacity. Specifically, it's

strength-endurance of the finger flexor muscles and power-endurance of the larger pulling muscles of the arms and torso that are being tested in doing long, sustained boulder problems and roped “resistance” climbs. The burning muscle pump is the hallmark of the anaerobic lactic (glycolytic) energy system being pushed to near its limit, and of course many redpoint attempts end accordingly, with a Hindenbergian forearm pump that makes climbing to (or clipping) the chains impossible.

In an attempt to improve anaerobic capacity, many climbers make strength-endurance of the fingers (and power-endurance of the pulling muscles) the primary focus of their training. While a few weeks of targeted anaerobic-capacity training will often bring favorable adaptations and some noticeable gains on the rock, many climbers train this energy system too frequently and too long without variation. It's interesting to note that climbing (or training) to the point of a blistering pump more than about two or three days per week over the long term often leads to decreasing endurance, possibly due to the damage of chronic acidosis to mitochondrial function and related enzymes. If you have ever engaged in frequent high-volume strength/power-endurance training, only to discover that you're pumping out even faster on the rock, you now know why. So what's the best strategy for training anaerobic capacity?

I've pointed out several times that the anaerobic lactic energy system is the least trainable of the three energy systems, and therefore making significant year-over-year gains in anaerobic capacity demands getting stronger (anaerobic alactic training) and increasing aerobic power and capacity. The best training approach, then, is to partake frequently in anaerobic alactic and local aerobic energy system mesocycles, and then to perform a two- to four-week mesocycle of pumpy anaerobic lactic training prior to a climbing road trip or competition. This approach will boost all three energy systems, prevent overtraining, and provide you with the

important anaerobic lactic system adaptations (cellular buffering and increased anaerobic enzymes) in the final weeks before your event.

With this understanding, you can think of your actual strength-endurance (and power-endurance, covered in chapter 9) workouts as mostly training “tolerance” to the acidosis and high lactate levels that result from pushing your anaerobic lactic system to the limit. This energy system is predominant in sustained, high-intensity exercise lasting fifteen seconds to about two minutes—thus each exercise presented in this section must be performed in a way that roughly meets this duration criteria . . . and gets you pumped!



SHORT-DURATION FINGERBOARD REPEATERS

Repeaters are the standard fingerboard exercise for developing strength-endurance and improving the ability to climb onward despite a building pump. There are many different repeater protocols in the public domain, but below I will outline what I feel are the two most effective approaches. First I’ll present short-duration repeaters, then I’ll outline the benefits of also training with long-duration repeaters.

Table 8.6 presents four difficult levels of short-duration fingerboard repeaters. While each level involves doing a set of six ten-second hangs, each higher level of difficulty affords a shorter rest between each hang. Each set of six hangs is

followed by a one-minute rest, after which you commence with the next set of six repeaters. Do up to five sets of six hangs. If doing these five sets with the Level 1 protocol isn’t difficult, then use the Level 2 protocol for the next workout. If you eventually come to find Level 4 training less than difficult, then you can add a ten-pound (or more) weight belt or use a pair of smaller holds

The beauty of short-duration repeaters is that the alternating periods of contracting and relaxing the forearm flexor muscles roughly approximates how we climb on the rock. The frequent rest periods allow blood flow to reoxygenate the working muscles and remove metabolic by-products—functional gains in the biological processes involved that will directly translate into improved performance on the rock!

A final distinction with regard to the grips you will train here: Research has determined that two primary muscles using in gripping the rock, the flexor digitorum profundus (FDP) and the flexor digitorum superficialis (FDS), contribute to force generation in crimp gripping at a ratio of 3:1, respectively. When open-hand gripping, however, force generation from these two muscle groups comes at nearly a 1:1 ratio (Vigouroux 2006). Given this knowledge, you can understand the importance of training both the crimp and open-hand grips; of course, you must always train the crimp position as a half crimp, *not* a full crimp with thumb lock. Applying this to repeater training, it would be a good idea

Table 8.6 Short-Duration Fingerboard Repeaters

Difficulty Level	Duration Hang/ Rest (seconds)	Number of Hangs per Set	Number of Sets	Rest Between Sets
Level 1	10/30	6	3–5	1 minute
Level 2	10/20	6	3–5	1 minute
Level 3	10/10	6	3–5	1 minute
Level 4	10/5	6	3–5	1 minute

to alternate between half-crimp and open-hand grips with each successive set performed.



LONG-DURATION FINGERBOARD REPEATERS

Long-duration repeaters are a bit less specific to how we generally climb, although they may roughly simulate the longer-duration forearm contractions common to shaking out at a mid-climb rest position. This point aside, the extraordinary benefits of long-duration repeaters are first, that longer-sustained contractions occlude blood flow—this ischemia results in an increasingly hypoxic and acidic cellular environment. Then, during the rest periods, forceful reperfusion (restoration of blood flow) develops throughout the forearm flexor muscles, as you recognize the “pump” feeling develop. This cycle of ischemia and reperfusion is an apparent stimulus for beneficial adaptations, although the mechanisms responsible are not fully understood. Furthermore, training that yields a reduction in intracellular pH (acidosis) will result in improved cellular buffering—an important adaptation in the final weeks of a training cycle before the pre-road trip or pre-competition training taper.

Once again, I am presenting you with four difficulty levels of long-duration repeaters in table 8.7. Begin with Level 1 and advance to the next level when you can complete five sets at the current level. Long-duration repeaters can become quite painful, so you may also benefit from mental

(pain tolerance) and central governor recalibration. **Warning:** Good fingerboard technique is essential for repeater training—maintain engaged shoulders



Fingerboard repeater training with guidance from the Zlagboard training app.
ZLAGBOARDUSA.COM

Table 8.7 Long-Duration Fingerboard Repeaters

Difficulty Level	Duration Hang/Rest (seconds)	Number of Hangs per Set	Number of Sets	Rest Between Sets
Level 1	30/30	4	3–5	1 minute
Level 2	30/15	4	3–5	1 minute
Level 3	60/30	4	3–5	1 minute
Level 4	60/15	4	3–5	1 minute

(chest out!) throughout each set, and end a hang early if you sense your shoulders beginning to loosen and shrug upward.



BOULDERING 4X4S

The bouldering 4x4 is a common indoor climbing method of training strength-endurance in the finger flexors and power-endurance in the larger pulling muscles of the arms and torso. Physiologically, 4x4s are kind of a climber’s equivalent of “gassers” sprint training performed by speed-endurance athletes such as soccer, hockey, and football players. This increasingly difficult workout trains both anaerobic capacity and aerobic power, if you follow the protocol outlined below. The short recovery periods are just as important as the brief spells of bouldering—so use a timing app or stopwatch to get it right.

There are many different ways of doing 4x4s, but Table 8.8 presents the most straightforward and my personal favorite—and it’s a great protocol to do with a training partner. The 4x4 involves doing four ascents (each) of four different boulder problems. Selecting the right problems is a bit tricky, as you want each to be hard enough to push you but not so hard you’ll fail (except, perhaps, on the fourth pass). Obviously, you’ll need to choose problems a few grades below your maximum

level—for example, if your current (this season) hardest boulder ascent is a V9, then you should select problems in the V4 to V6 range for your 4x4s. For the purpose of finger training, it would be best to select boulders with smallish holds, but void of any big stopper reaches. Climbing each problem should only take thirty seconds, so don’t pick long and involved problems or any that require stopping to rest. The training protocol is this: You have one minute for each boulder ascent (including post-climb rest), and then you’ll take four minutes of rest after every fourth climb. So beginning on the first problem, let’s say it takes twenty-five seconds to ascend; you’ll then rest for the remainder of the first minute (thirty-five seconds). Do a second, third, and fourth lap on the first boulder problem using the same climb-rest scheme that adds up to exactly one minute. Upon completing the fourth ascent (actually, at the end of the fourth full minute), take four minutes of complete rest—if you’re training with a partner, you can now time her set of four climbs. Next, you’ll repeat this process on the second of your four chosen boulder problems. Again, you have one full minute of climb-rest time per lap, after which you get four minutes of rest. Repeat this process with the third and fourth boulder problems. All totaled this 4x4 protocol takes about thirty minutes. This is a powerful anaerobic endurance

Table 8.8 Bouldering 4x4 Protocol

	Boulder 1		Boulder 2		Boulder 3		Boulder 4	
	Climb	Rest	Climb	Rest	Climb	Rest	Climb	Rest
Ascent 1	~30"	~30"	~30"	~30"	~30"	~30"	~30"	~30"
Ascent 2	~30"	~30"	~30"	~30"	~30"	~30"	~30"	~30"
Ascent 3	~30"	~30"	~30"	~30"	~30"	~30"	~30"	~30"
Ascent 4	~30"	~30"	~30"	~30"	~30"	~30"	~30"	~30"
Rest	4 minutes		4 minutes		4 minutes		4 minutes	

training technique—so get a partner, use a smart-phone timing App, and get pumped!



HYPERGRAVITY ISOLATION TRAINING—STRENGTH-ENDURANCE PROTOCOL

Performed exactly as described in the HIT System Maximum-Strength Protocol (page 174), here you will be using lighter weights and climb for a much longer duration to train strength/power-endurance (anaerobic capacity). The acceptable extremes for duration of each set are twenty seconds and two minutes, although I believe thirty to ninety seconds is ideal (see table 8.9). Consider taping your fingers (figure-8 method) for skin protection so that you can push hard throughout each set, rather than needing to terminate early due to skin pain. This very grueling workout creates significant fatigue—full recovery may take up to seventy-two hours, so plan accordingly.



SYSTEM WALL “REPEATERS”

A System Wall is a specialized training platform with a large assortment of differing holds organized

in a way that each half of the wall is a mirror image of the other. System Walls typically overhang anywhere from 20 to 45 degrees past vertical, and they are often only six to eight feet wide and nine to, at most, fifteen feet high. The utility of a System Wall is the ability to train a specific finger grip, arm position, movement, and body position in a very precise and repeatable way for both arms and sides of your body.

Common uses of a System Wall include working undercling arm strength and reaches, one-arm lock-offs and hand bumping, deadpoint moves, compression moves, and twist-lock moves, among many other possibilities. A well-outfitted System Wall with dozens of different holds can offer a really good targeted workout of the forearm (gripping) muscles, the larger pulling muscles of the arms and torso, and the many core muscles of the torso.

In terms of training strength-endurance of the forearm flexors (and power-endurance of the pulling muscles), you’ll want to use a protocol very similar to the HIT System strength-endurance protocol outlined in Table 8.9. Initially it will take a little experimentation, but develop four to six different

Table 8.9 HIT System—Strength-Endurance Protocol

Grip Position	Weight Used HIT “Novice”	Weight Used HIT “Advanced”	Set Duration (reps/set)	Number of Sets	Rest Between Sets
2-Finger “3rd Team”	None	5 lbs.	20–40 secs.	1–3	3 mins.
Pinch	None	5 lbs.	20–40 secs.	1–3	3 mins.
2-Finger “2nd Team”	None	10 lbs.	30–60 secs.	1–3	3 mins.
2-Finger “1st Team”	None	10 lbs.	30–60 secs.	1–3	3 mins.
Half Crimp	None	20 lbs.	45–90 secs.	1–3	3 mins.
Open Hand	None	20 lbs.	45–90 secs.	1–3	3 mins.

* Weights are approximations for a 160-pound climber. Use similar percentages of your body weight.

The History of Hypergravity Isolation Training (i.e., HIT System Training)

In the mid-1990s I set out to develop a climbing-specific method for training functional strength in the important finger and arm flexor muscles, as well as the vital muscles of the shoulders and core. Much experimentation with equipment designs and training protocols led to what I call Hypergravity Isolation Training (HIT). While similar to the more variable System Wall Training, HIT workouts are highly focused on training specific grip positions while pulling and twisting up a severely overhanging wall. Since I first promoted this exciting training method in my book *How to Climb 5.12* in 1997, HIT workouts have been utilized by thousands of climbers around the world. The effectiveness of HIT (not to be confused with the popular weight lifters' "HIT workout") is a result of its fulfillment of the four fundamental requisites for training grip strength detailed on page 167.

The adjacent photo depicts me training on the third-generation HIT Strips and HIT Pinches (available at www.Nicros.com), a unique platform I developed for optimal HIT workouts. Each HIT Strip possesses identical crimp edges and two-finger pockets that are ideal for laddering up and down until failure, although in lieu of the HIT Strips you can also arrange pairs of identical crimp, pocket, and pinch holds to train on. The ideal wall angle for HIT training is 45 to 55 degrees past vertical. Weight added around the waist is increased or decreased to produce near failure of the finger flexors in the required amount of time—fifteen seconds if using the HIT maximum-strength protocol and thirty to ninety seconds when using the HIT strength/power-endurance protocol. Here are a few more tips for effective HIT System training.

- Use a snug-fitting weight belt(s) or a weight vest, and adjust the weight for each grip position. Record weights used in a training notebook.
- Climb briskly and without hesitation—no stopping or chalking mid-set. Always step off the wall as opposed to risking an uncontrolled fall while climbing with added weight.

- Climb with normal foot movements and body turns. Small- to medium-size footholds are best, because too much thought on footwork will slow you down.
- Rest breaks between sets must be exactly two or three minutes; more advanced climbers should use the former. Use a stopwatch and stick to the planned order and schedule of exercises.
- Keep a training notebook in which you log each set, weight added, the number of reps (hand moves) performed, and the length of rest intervals. This way you'll always know what weight you need for a given set, and you can quantify and track your gains in finger strength!
- Tape your fingers using the Figure-8 method (page 286) to reduce skin pain, especially when training with high weight loads.
- Visit www.Training4Climbing.com for more information on building a dedicated HIT System Wall.



The author (plus twenty pounds) on a HIT System.



System Wall Repeaters isolating the same movement on both sides. KRIS HAMPTON COLLECTION

“theme repeaters” that you can climb up and down through—using identical left and right hand/arm positions—for between thirty and ninety seconds without stopping. Since System Walls are short, you’ll likely need to lap up and down through the sequence several times to reach the desired training time and level of fatigue. On a less steep System Wall (closer to just 20 degrees past vertical), you may be able to climb for sixty to ninety seconds for each set, whereas a steeper wall may limit your training set to just twenty to forty seconds. Either way, take a three-minute rest between each set.

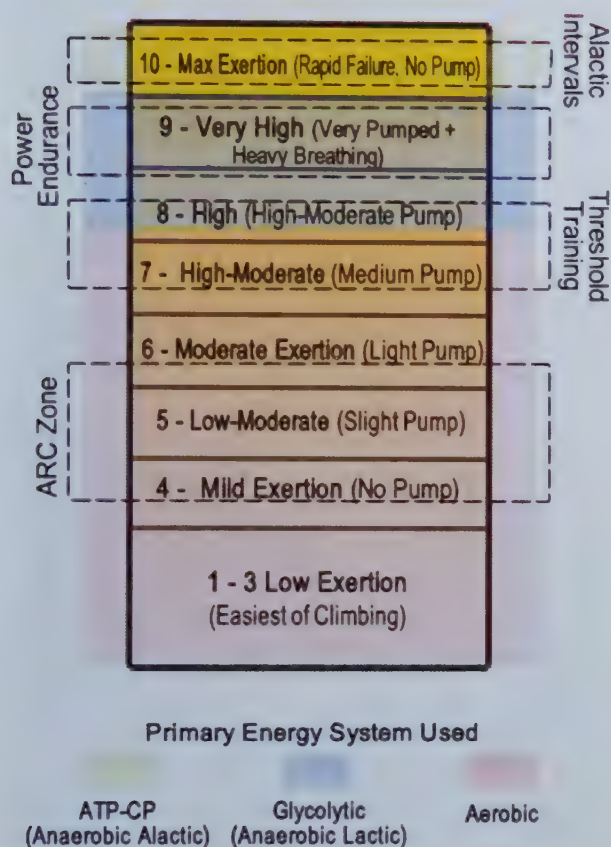
Do one or two sets on each of four to six different “theme repeaters” for a total of four to twelve sets in aggregate. Long term, you can increase difficulty by using smaller holds, larger arm moves, and more difficult and strenuous body positions, as

well as by adding a weight belt. In chapter 9 we’ll examine using a System Wall to train arm and core strength in a variety of arm and body positions.

Training Forearm Local Aerobic Endurance

Effective training of the aerobic energy pathway via climbing is possible only if you set aside the desire to climb for performance (near your limit) and instead embrace the potentially boring process of climbing high volumes of moderate rock (or plastic). No matter if the climbing is done indoors or outside, this approach will naturally involve lots of submaximal climbing at varying intensities. The critical guideline to obey is that the climbing intensity never ascends significantly into the anaerobic

Figure 8.3 Training Zones for Climbers



Effective climbing-specific aerobic training must be performed between 5 and 8.5 on the rating of perceived exertion.

zone, the hallmark of which is a deep muscle pump and shortness of breath. Ultimately, you want to find the margin of the anaerobic lactic zone and strive to climb mostly just below it (in the aerobic zone), only occasionally crossing into the low end of the anaerobic zone. A light to, at times, moderate forearm pump is fine and desirable; however, a flaming pump, heavy breathing, and a growing sense

of losing control are clear signs you've climbed too deep into the anaerobic zone.

Self-control and a strong intention to climb mainly in the aerobic training zone are essential to properly training this energy pathway. By keeping your perceived exertion and exercise intensity between 5 and 8.5 on a scale of 1 to 10 (see figure 8.3), you can rest assured that you're getting it right. Detailed below are five methods of training local aerobic endurance.



ARC TRAVERSES ("RECOVERY CLIMBING")

ARC is an acronym for aerobic, recovery, and capillary training. The goal of ARC training is to enhance recuperative blood flow via an extended period of lower-intensity climbing. Such a training protocol—if you can stick to it—singularly targets the aerobic energy system and therefore yields little (or no) muscle pump. Done right, this light climbing workout will generate no fatigue to recover from and, in fact, will help accelerate recovery from the previous day's intense workout. The problem, however, is that the typical enthusiastic climber has a tough time visiting a climbing gym and refraining entirely from doing more difficult routes or boulder problems, but even a small sampling of more challenging routes or boulders will sink the ARC.

Rather than calling it ARC training, I prefer the term "recovery climbing," since this unambiguously expresses the actual mission of the activity: Easy climbing that will enhance recovery rather than get in the way of it. The modus operandi we are after is similar to that of a runner going for a "recovery run"—the climbing (or running) pace must be so casual that you can hold a conversation throughout. On a perceived exertion/intensity scale of 1 to 10, the ideal recovery-training zone is between a 4 and 6 (see figure 8.3); go any higher and it's no longer an ARC/recovery session.

Recent Research on the Importance of Local Aerobic Endurance Training of the Finger Flexors

Strength and endurance of the finger flexors is now considered to be one of the key determinants of sport climbing performance. Current studies have focused on understanding the open-crimp position on plastic holds. When sport climbers are placed in groups characterized by on-sight and redpoint abilities, several significant physiological differences have been found. It has now become evident that these differences are training adaptations that affect the muscle structure of the forearm flexors, and these may influence the way we ascend and rest on a route.

Given our understanding of rock climbing physiology, there are some training modalities that are likely to enhance sport climbing performance—for example, climbing-specific training that focuses on improving the aerobic capacity of the forearm flexors (not to be confused with training whole body aerobic capacity ($\text{VO}_2 \text{ max}$), such as running). Although we recognize that $\text{VO}_2 \text{ max}$ is important for sport climbing, the aerobic capacity that is specific to the forearm flexors appears to be more important. Our laboratory has found that after sustained and intermittent contractions to failure, higher-level sport climbers have a greater forearm aerobic capacity, as they can off-load and recuperate oxygen faster and to a

greater extent than lower-level sport climbers. Interestingly, the speed of this oxygen recovery has been strongly correlated to creatine phosphate resynthesis (replenishing our short-lived but powerful anaerobic energy system). As such it is important to include training that stresses the aerobic metabolism in the forearms—that is, a high number of repetitions at a relatively low intensity, as this is likely to improve the ability to recover between holds on a route.

Based on the relationship between sport climbing ability and oxygen recovery (and creatine phosphate resynthesis), the physiological benefits of resting on a route are going to be of paramount importance to a successful ascent. This would be particularly important for sections on a route that come just before the crux move. Further, as we have found that the nondominant finger flexor muscles recover oxygen slower compared to the dominant, it will be worth spending a little extra time resting (and training) the nondominant side to ensure maximal performance. This will be particularly important if the route requires a lot of workload to be placed on the nondominant forearm.

—Dr. Simon Fryer, School of Sport and Exercise,
University of Gloucestershire, UK

For the purpose of seriously developing the aerobic energy system, however, the ARC/recovery climbing protocol misses the mark—it's just too low-intensity. Therefore, save this type of recovery training for occasional use the day after a hard maximum-strength or strength-endurance workout. Traversing along the base of a large indoor climbing wall, using mainly large holds, is the easiest way to go about this—without the tether of a belay you can easily step off the wall at the first sign of a developing pump. Two common approaches are to

simply traverse around the gym until you get bored, or to do a series of five-minute traverses with two- to five-minute rests in between. Aim for a total climbing time of around thirty minutes, and don't get pumped!



FINGERBOARD MOVING HANGS

If you do a lot of your training at home by way of a fingerboard, then the “moving hang” is your ticket to getting a decent local aerobic endurance



Fingerboard “moving hang” with feet on the wall.

workout. Yes, this is an exceedingly boring way to train, and you should take every opportunity to train at a climbing gym instead. Still, a motivated climber can do a lot of beneficial training at home, including antagonist training, mobility work, generalized aerobic training, and fingerboard training, the latter of which is as specific as home training gets outside of having a climbing wall.

As the name implies, moving hangs involve circulating your hands around the board in a semi-continuous fashion for several minutes while

keeping your feet fixed on a chair (or footholds fastened to the wall) to de-load your body and reduce the intensity as needed. It helps to visualize being on an actual rock climb as you hang, pinch, pull, and cross-through in a climbing-like way. When you start to get pumped, move onto the jug holds for a brief shakeout before proceeding with more “climbing.” Continue in this way for three to six minutes, being sure to stay in the aerobic training zone of between 5 and 8.5 (figure 8.3). Do this right and you will get a moderate pump, but not a deep, painful pump (a sign you’re training anaerobic capacity, not the aerobic system). Do three to six moving hangs with a climb-rest ratio of between 2:1 and 1:1. Consider doing some core or antagonist training during the rest periods.



CAMPUS BOARD “HAND PLAY”

This exercise is basically the same as the moving hangs described above except that it’s performed on a campus board. It’s important to note that this is a feet-on campus exercise, so you’ll need to place your feet on a chair or on wooden strips or climbing holds bolted onto the wall below. It’s important to recognize that this is not a dynamic exercise, so it’s in no way like the campus board exercises used to develop contact strength. The goal here is aerobic endurance training by way of easy to moderate hand movements around the board and liberal use of shakeouts from the largest holds available. Depending how much body weight is shifted onto the feet, you should be able to “hand play” around the campus board for at least three to five minutes. Keep the pump under control, and step off the board before attaining a deep, painful pump—a sign of cellular acidosis as a result of anaerobic energy production becoming predominant. Err on the side of too little pump and shorter sets, rather than developing deep fatigue and the need for longer rests. Do four to six sets with nearly equal periods of climbing and resting.



THRESHOLD INTERVALS ON A TREADWALL OR BOULDERING WALL

A Treadwall is an expensive piece of equipment, but to a climber-for-life type (like me) it's worth every penny. Depending on the size of holds used and the amount of weight added via weight belt (if any), a Treadwall can be effectively used to train anaerobic and aerobic power (via brief, intense intervals lasting less than twenty seconds), anaerobic capacity and the lactic system (hard, pumpy intervals lasting thirty seconds to two minutes), or the aerobic energy system by way of longer (three to six minutes) moderate-intensity intervals with equal or shorter rest periods. The focus here, of course, is the latter application of training local aerobic capacity. If you don't have access to a Treadwall, you can also execute the following training protocol on a bouldering wall.

As with all the previous exercises in this section, the fundamental requirement is to climb mainly in the aerobic training zone—that is, between 5 and 8.5 (see figure 8.3). A steep Treadwall or bouldering wall (more than 30 degrees past vertical) may make this a challenging endeavor, however, in which case you'll need to use large, juggy holds and frequent shakeouts. End a climbing interval early (say at three minutes) rather than push deeply into the anaerobic zone—if you get a painful pump, experience shortness of breath, and need to fight to stay on the wall, then you're well beyond the target training zone. If there's one liability to owning a Treadwall or home bouldering wall, it's the ease of use (and overuse) and the difficult-to-resist tendency to always climb into the anaerobic zone. Climbers who succumb in this way inevitably end up getting lackluster results (despite all their hard training) and often end up victims of overtraining syndrome.

As you can probably surmise by now, the threshold interval protocol is to climb a series of moderate-intensity intervals lasting three to five minutes. The ideal training intensity is between a 7 and 8.5 out of 10 on the RPE scale—this puts you



Threshold interval training on a Brewer's Ledge Treadwall.

right at the anaerobic threshold, where the aerobic system is taxed to its limit. Do four to eight intervals with a 1:1 climb-to-rest ratio. Err on the side of too little pump and only a few sets rather than push it too hard, for too long, and acquire too deep of a pump.



ROUTE CLIMBING INTERVALS

This is by far the best local aerobic training protocol if you're lucky enough to have membership at a large climbing gym. You will need to exert self-control in not getting drawn into climbing for performance—the goal here, as with all the aerobic training exercises, is to engage in sustained climbing

that yields just a light to moderate pump. This is an excellent partner exercise, so enlist someone who also understands the value of doing a high volume of moderate climbing. Take equal turns climbing and belaying and you'll have a nice 1:1 climb-to-rest ratio.

To maintain a proper intensity level, choose routes that are two to four number grades below your limit. Accordingly, if you are a 5.12 redpoint climber it's best to train only on routes in the 5.8 to 5.10+ range—yes, these routes will seem really easy, but the training goal is to climb lots of volume

and never get more than a moderate pump. Check your hard-climber ego at the door and assume a new mind-set and mission—to climb three times as many pitches as anyone else in the gym! To do this, you'll likely need to climb twenty or more top rope or lead climbs. By keeping the routes easy and the pump "light," both you and your partner should be able to complete between six and twelve routes per hour, depending on length. If you develop a deep muscular pump or labored breathing, immediately move onto easier routes rather than allow your training focus to shift into the anaerobic zone.

Daniel Woods on the first ascent of one of the world's hardest boulder problems, The Process (V16), Buttermilks, California. DAVID CLIFFORD

